When Sampling Shapes the Story: Representation and Claim Reliability in Historical isiXhosa NLP

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Abstract

Sampling is often treated as a neutral preprocessing step in historical-language NLP. We test whether it instead changes both *model outcomes* and *the conclusions researchers draw from texts*. In a controlled isiXhosa-newspaper study, fixed-budget BPE and Unigram tokenisers show improved English-view segmentation but less compact isiXhosa-view segmentation. We then introduce *ArchiveLens*, an auditable workflow that tests historical surface-pattern trends under alternative sampling units, permutation calibration and repeated-text diagnostics. Although 23/120 Imvo patterns reverse estimated direction, the count is not unusual under permutation; 29 exhibit Holm-adjusted estimator-gap magnitude sensitivity. We argue that responsible historical NLP requires *exposure-aware evaluation and calibrated claim warnings*, not assertions of cultural change from surface patterns.
Keywords: Responsible NLP; historical isiXhosa; sampling sensitivity; measurement validity.

Introduction and Related Work

The *same* historical newspaper text can yield different numerical descriptions depending on whether researchers count pages or words: in matched *Imvo Zabantsundu* years, the archive's English-view share is *22.64% by equal page* versus *29.47% by word mass*. Neither is necessarily incorrect; each answers a different exposure question. Such choices matter especially when working with the isiXhosa Intellectual Traditions archive, whose digitised newspapers preserve historically important African writing without constituting a representative sample of speakers or beliefs (Schoots, Ngwendu, De Wet, & Muftic, 2023).

Multilingual tokenisation can impose unequal computational costs (Petrov, La Malfa, Torr, & Bibi, 2023), and its intrinsic efficiency need not predict downstream performance (Ali et al., 2024). Historical collection audits highlight the influence of source composition (Beelen, Lawrence, McDonough, & Wilson, 2025), while data statements demand explicit populations and limitations (Bender & Friedman, 2018). Building on behavioural stress testing (Ribeiro, Wu, Guestrin, & Singh, 2020), we ask two connected but *distinct* questions: **(Q1)** Can controlled sampling shift computational representation across archive views? **(Q2)** Can sampling choices also change the apparent *descriptive trends* measured in historical text? We test Q1 with a factorial intervention and Q2 with our *ArchiveLens* auditing instrument; we do not infer that tokenisers caused historical trends.

Methods

Q1—intervene on model exposure. In *Imvo*, a 2×2 design varies (A) page- versus word-weighted allocation between English and isiXhosa archive views, and (B) page- versus word-weighted choice *within* views (PP, PW, WP, WW). Across BPE and Unigram, four arms and five seeds, we fit 40 principal tokenisers (2,000-piece vocabulary), each on $\sim$500,000 alphabetic characters. Training uses 1885 and Jan–Sep 1887, validation Oct–Dec 1887 and evaluation 1888; 1889–94 is a diagnostic. We assess pieces per 100 alphabetic characters and smoothed count-bigram bits per character (BPC). Factorial contrasts are paired and average over the other factor; uncertainty resamples publication issues.

Q2—audit claims about the archive. ArchiveLens uses *Imvo* and *Isigidimi samaXosa*, selecting 120 common within-word character four-grams by frequency in *1887 only*. For 1888 minus 1885, it contrasts character-position mass, equal-page, equal-issue and equal-month rates (per 10,000 eligible four-gram positions). A 3,000-draw month/issue bootstrap describes uncertainty. We then calibrate direction-reversal counts and estimator-gap magnitudes using 4,999 within-month, whole-issue year-label permutations; Holm adjustment tests 120 patterns per source and statistic family. Leave-one-issue-out influence and exact cross-issue 13-word repetition masking probe sensitivity. These *post-hoc conditional* analyses use previously inspected years, not independent external confirmation.

Results

Sampling changes what the model learns (Fig. 1A). Changing allocation lowers English-view BPE segmentation cost by 1.878 pieces/100 characters (signed $\Delta = -1.878$, 95% issue CI $[-1.972, -1.782]$), but raises isiXhosa-view cost by 0.453 (CI $[0.407, 0.497]$). Unigram shows the same directional trade-off (-1.795 EN; $+0.555$ XH). *The gain is not language-view-neutral.*

Model utility is a second decision (Fig. 1B). For English-view BPE, allocation improves bigram loss by -0.0372 BPC (95% issue CI $[-0.0424, -0.0321]$), while within-view page choice worsens it by $+0.0194$ BPC (CI $[0.0115, 0.0277]$). The combined policy improves 1888 BPC (-0.0178) but is *inconclusive* in 1889–94 ($+0.0045$, six-year CI $[-0.0027, 0.0131]$). Notably, isiXhosa-view BPE segmentation gets less compact while its average BPC improves slightly (-0.00275): intrinsic compactness is not a sufficient downstream proxy.

Audit the warning itself (Fig. 1C). Under alternative estimands, 23/120 *Imvo* and 4/120 *Isigidimi* four-gram trends reverse point-estimate sign, but none shows supported opposite directions under both estimands. Imvo's reversal *count* is

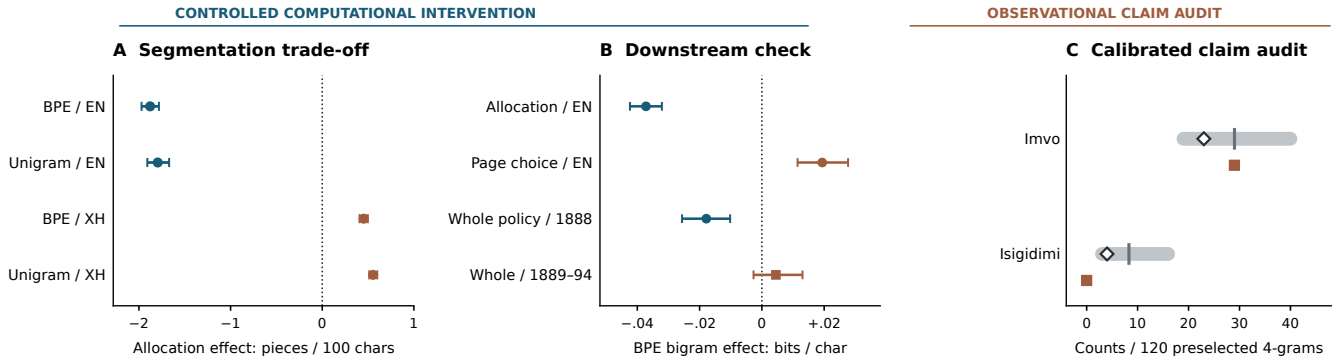


Figure 1: Two linked Responsible NLP failure modes. A: 1888 allocation effects on segmentation for archive English (EN) and isiXhosa (XH) views; points and lines show 95% issue-bootstrap intervals (negative = more compact). **B:** BPE English-view count-bigram effects for allocation, within-view page choice and combined policy (1888 issue intervals; 1889–94 whole-policy effect has a six-year interval). **C:** ArchiveLens's observed sign-reversal counts (hollow diamonds) versus whole-issue permutation means (grey ticks) and 95% reference ranges (grey lines); coloured squares show *different* Holm-adjusted tests of estimator-gap *magnitude*. A sign reversal alone is not evidence of a historical semantic change.

not unusual under 4,999 permutations (23 observed; null mean 29.0; upper-tail $p = .888$). In contrast, *estimator-gap magnitude* survives Holm correction for 29/120 Imvo patterns (0/120 Isigidimi). Removing one issue reverses the mass-weighted sign in 7/32 flagged Imvo patterns; masking exact repeated passages eliminates 10 of Imvo's original 23 reversal warnings, while changing sample eligibility.

Discussion: Responsible NLP

An algorithm can optimise the wrong denominator. A language-averaged segmentation score hides the empirically opposed view-specific effects. Even a per-view segmentation gain cannot replace task-level measurement: allocation and page-choice effects oppose one another for English bigram loss, and isiXhosa BPC can move differently from compactness. Responsible evaluation must state the target exposure distribution *before* declaring one policy better.

A warning is not a discovery. ArchiveLens makes the analysis denominator visible and tests the reliability of its own alerts. The negative result—no excess of raw reversal counts under the permutation reference—guards against exaggeration. Corrected estimator-gap tests nevertheless find measurable sensitivity in Imvo, while issue influence and repeated text explain why additional checks are needed. They do not identify historical causes; removing duplicated material alters the evaluated population. Isigidimi contains fewer issues (12 in 1888 versus Imvo's 52), so zero Holm-positive gaps must not be construed as a more reliable or fairer archive.

From warning to accountable practice. Instead of assigning an unexplained "fairness score", ArchiveLens records each tested statement's source and dates, its denominator and alternative estimands, issue-based uncertainty, calibration reference, and influential or repeated text. A defensible reporting rule is to describe a result as *stable under tested assumptions*, *measurement-sensitive*, or *unresolved*; none of these labels certifies cultural truth. This matters when an apparently precise estimate might later inform search, access to collections or interpretations of political writing.

Costs of misinterpretation. Treating a short character sequence as a historical concept, or an archive view as an individual's language, could misrepresent the people represented by the collection. Our audit can flag this technical risk but cannot resolve it. Further work needs language experts and archivists to validate meaningful word forms, editorial context and intended applications, alongside independent corpora and community-grounded assessment.

Limits and responsible use. `EnTxt/XhTxt` are archive views, not human-validated word-language identities. The files are corrected transcriptions, not raw-OCR confidence annotations; genre, editorial practice, preservation and earlier analyst exposure to test years remain confounders. Four-grams are surface strings, not morphemes or cultural ideas, and both newspapers share one source collection. We neither measure societal fairness harm nor establish historical semantic change. Human linguistic/archival review and genuinely external evaluation are needed before those claims.

Conclusion

Which texts are exposed, which outcome is optimised, and which claims survive counterfactual weighting are inseparable parts of responsible historical NLP. Our controlled study and calibrated ArchiveLens workflow make these choices empirically visible and reproducible. Source, tests, frozen protocols and full evidence are in the open research repository.

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